

Total No. of Questions : 6]

SEAT No. :

P3702

[Total No. of Pages : 2

Engg. - 49

T.E. (Computer Engineering) (Semester - I)

Database Management Systems Applications (In Sem.)

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Draw the E-R diagram which model University Database. **[4]**

- i) List the entity sets and their primary keys.
- ii) Extend the E-R diagram, using aggregation, to model the case where we want to record evaluations of a student by a guide on a project

b) Explain View and Index in SQL with suitable example. **[4]**

c) Consider relational schema **Employee** (*Empno*, *EName*, *DeptNo*, *Salary*), **Department**(*DeptNo*, *DName*) **Write SQL Queries for following questions (Any two)** **[2]**

- i) List Employee Names of 'Computer' Department.
- ii) Find average salary of each department.
- iii) Find Department name of employee name 'Amit'

OR

Q2) a) Consider the following Relations. It defines the schema of the database application for a bank. It manages the branches and customers of the bank. Customers take loans (borrow money) or open accounts (deposit money) at one or more branches. **[4]**

Branch (*B_No*, *B_name*, *B_city*, *asset*) , **Customer** (*C_No*, *C_Name*, *C_city*, *Street*)

Loan (*Loan_no*, *B_name*, *amount*) , **Account** (*Acc_No*, *B_name*, *Balance*)

Borrower (*C_No*, *Loan_No*), **Depositor** (*C_No*, *Acc_No*)

P.T.O.

Write SQL Queries for following questions (Any two)

- i) Find loan data, ordered by decreasing amounts, then increasing loan numbers.
- ii) Find the pairs of names of different customers who live at the same address but have accounts at different branches.
- iii) Find the names and address of customers who have a loan for an amount exceeding 3 times their current balance.
- b) Explain 1NF & 2NF (Normal Form) with suitable example. [4]
- c) List Advantages of a DBMS over file-processing systems. [2]

- Q3)**
- a) Explain CAP theorem and BASE properties in NoSQL Database with suitable example. [4]
 - b) Consider schema User (user_id, age, status). Write MongoDB Schema statements for following queries (**Any four**) : [4]
 - i) Create Collection and Document
 - ii) Insert Data and Update Document
 - iii) Find all the users whose age is equal to 50 or status is “A”
 - iv) Update the user’s age increment by 3 whose status is “A”
 - v) Delete the users whose status is “A”
 - c) List difference between Relational Database and NoSQL Database. [2]

OR

- Q4)**
- a) Explain Aggregation using MongoDB with suitable example. [4]
 - b) List different NoSQL Data Models. Explain document based NoSQL data model in short. [4]
 - c) Explain sharding (Horizontal Scaling) in MongoDB. [2]
- Q5)**
- a) Explain two phase locking protocol with suitable example. [4]
 - b) What is Serializability? Explain Conflict Serializability with suitable example. [4]
 - c) Explain Performance Tuning in NoSQL in short. [2]

OR

- Q6)**
- a) Explain Timestamp based and log based concurrency control protocol. [4]
 - b) Explain Query Optimization in Relational Database. [4]
 - c) Explain different Database Transaction States. [2]

